Division Using Binary Search

Problem Submissions Leaderboard

RATE THIS CHALLENGE

Given two integers dividend and divisor, divide them without using the multiplication (*), division (/), or modulus (%) operators.

Return the quotient obtained after dividing the dividend by the divisor.

Input Format

The first line contains an integer dividend. The second line contains an integer divisor.

Constraints

-2^31 ≤ dividend, divisor ≤ 2^31 - 1 divisor ≠ 0

Output Format

Print the quotient.

Sample Input 0

```
10
3
```

Sample Output 0

```
3
```

Change Theme Language Python 3

```python
import sys

def divide(dividend: int, divisor: int) -> int:
    INT_MAX = 2**31 - 1
    INT_MIN = -2**31

    if dividend == INT_MIN and divisor == -1:
        return INT_MAX

    negative = (dividend < 0) ^ (divisor < 0)

    a, b = abs(dividend), abs(divisor)
    quotient = 0

    for i in range(31, -1, -1):
        if (a >> i) >= b:
            quotient += 1 << i
            a -= b << i

    return -quotient if negative else quotient

```

```
if __name__ == '__main__':
    input_data = sys.stdin.read().split()
    if input_data:
        dividend = int(input_data[0])
        divisor = int(input_data[1])
        print(divide(dividend, divisor))

```

Line: 28 Col: 1

Upload Code as File

Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

Download

```
10
3
```

Your Output (stdout)

```
3
```

Expected Output

Download

```
3
```